
Reference, Retrieve, Resolve: How Small a Language Model Can Read a Reference in Context¹

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Abstract

On-device assistants mostly do one shape of task: a reference in context, a short question, a lookup and a rule. We build a narrow eval of that shape and ask how small a model can do it. The reference is the Generation I Pokémon type chart and the typing of all 151 species; the answer is the damage multiplier, a six-way choice with a code-generated key. Nine open models from 1B to 235B and one closed ceiling model run a pinned 400-item set at three epochs, one host per model, with a pre-registered second draw of 400 and two controls. Four findings. On this ladder no model at 8B or below that answers in one pass reads the chart as a grid; the cleanest step is within one family, Gemma 3 from 0.28 at 4B to 0.51 at 12B. Rendering the chart as one line per attacking type raises nine of ten models, by up to 0.18, up to a rung. A reasoning trace is worth more than any rung: a thinking 8B (0.850) beats every one-pass model on the ladder including a 235B mixture, and parameter counts did not place the two mixtures. And the score is reading, not memory: recall alone is 0.14 to 0.28 through 27B, and relabelling the fifteen types, which makes memory useless, moves no model by more than 0.06 at a matched token budget. Two swings that looked like capability were harness artefacts. Harness, key, item sets, registration files and ledger are released.

1 Introduction

Broad capability evals become trivia, and the number stops meaning one thing. This paper is one narrow, procedural task, a key nobody typed, and the question a deployment engineer asks: *what is the smallest model that can do this?*

The task shape is the one on-device models do all day. Apple’s on-device foundation model is about 3B parameters [Gunter et al., 2024], and sub-billion models are studied for the same seat [Liu et al., 2024]. A reference sits in context (a calendar, a settings table, a recipe), the user asks something short, and the model has to find the right line, read one or two facts, and apply a rule. We instantiate it with the Generation I Pokémon type chart, a 15×15 table of multipliers, and the typing of 151 species, both given in the system prompt. The question is “A Ground-type move hits Golem. What is the damage multiplier?” and the answer is one of $\{0, 1/4, 1/2, 1, 2, 4\}$: find Golem in the list (Rock/Ground), read two chart cells, multiply. No Pokémon knowledge is required or rewarded; a model that has never heard of Golem can score 100%.

Two properties of the domain are features. The key is derived by code from public tables, so every one of 2,265 cells is known exactly and the scorer can be shown to fail on a planted wrong answer. And the Generation I chart differs from the chart every model has memorised on four cells

¹Code, data, the seven registration files with their hashes, every run log, and the defect ledger: <https://github.com/collapseindex/pokemon-gen1-eval>. Total API spend for everything reported here: \$12.05.

(Bug/Poison in both directions, Ghost→Psychic, Ice→Fire), so the reference contradicts the prior on 71 of the 400 items. A model that follows the given chart against its own memory will follow your calendar.

Our contributions.

1. **A pre-registered, replicated size ladder on reference-following (§5).** With the reference as a grid, no non-thinking model at 8B or below reads it; a thinking 8B does (§5). The one step free of host, quantisation and family confounds is Gemma 3 from 0.28 at 4B to 0.51 at 12B; the cross-family order agrees on two disjoint item sets but carries those confounds; Gemma at 4B and Llama at 8B are not distinguishable. Two controls, no reference and a relabelled reference, show the score is reading at every rung that scores above chance (§5).
2. **Active parameters did not place the mixtures (§5).** A 30B mixture-of-experts with 3.3B active scores level with a 27B dense model on both item sets (paired over items, $+0.026$ $[-0.014, +0.068]$) and 0.19 below the same-family 32B dense model; the 235B mixture also sits below that 32B on two hosts. Both orderings against the 32B carry a thinking confound: it reasons on every call and neither mixture does.
3. **The rendering of the reference is a first-order effect (§5).** The same 225 cells as one line per attacking type raise nine of ten models, by up to 0.18, and move the knee to about 4B on point estimates.
4. **An error taxonomy on the misses (Table 3).** Arithmetic is at most a small part of the failure at any size; immunity misses change from random to mirror reads with size; mirrored answers peak in the middle of the ladder, where a model can read the grid and its memory can still beat it.
5. **A defect ledger that changed two numbers (§6).** A token budget moved one model by 0.13 and an upstream host moved another by a third of its calls; both would have been reported as capability without a scorer that keeps “no answer” separate from “wrong”.

Every registered prediction is in Table 4 with what happened to it: of thirty scored, twelve held, nine held in part and nine failed outright, and the nine that failed (Appendix A) are the paper’s most useful sentences.

2 Related work

Locating a capability threshold on one bounded task. The closest design to ours is [Chauhan and Pendyala \[2026\]](#), who sweep 15 open-weight models, 0.35B to 35B, on one fault-isolation task in a load balancer and report “a capability threshold near 3B active parameters”. Our result runs the other way: a 3.3B-active mixture behaves like a 30B model (§5). [Xu et al. \[2026\]](#) build RULER (not the long-context benchmark of the same name, [Hsieh et al., 2024](#), whose synthetic retrieval over long inputs is the nearer relative of a 3,000-token lookup): verified questions from expert emergency-response rules, scoring rule memorisation and rule application separately. That is the construct we measure, with a table for rules and a size sweep instead of a method sweep.

Knees, formats, harnesses. Loss scales smoothly [[Kaplan et al., 2020](#)]; task accuracy sometimes does not [[Wei et al., 2022a](#)], and [Schaeffer et al. \[2023\]](#) showed that a discontinuous metric can manufacture the knee. We report an exact and a graded score on the same items for that reason and find the knee in both; the graded score exceeds the exact one by at most 0.07. For mixtures, [Fedus et al. \[2022\]](#) decouple parameters from compute per token and [Clark et al. \[2022\]](#) show routed models scale along an effective count that is neither active nor total; our two MoE points are consistent with that. [Sclar et al. \[2024\]](#) show spurious formatting choices move accuracy by tens of points and [Sui et al. \[2024\]](#) that table serialisation changes what is read; our table-versus-rows result (§5) is the same phenomenon with an error analysis that says which misses the format removed. [Zheng et al. \[2024\]](#) and [Pezeshkpour and Hruschka \[2024\]](#) show option-order sensitivity; we hold the order fixed and report the predicted-letter distribution, and run one shuffled control (§5). Treating the harness as a source of error in its own right follows [Biderman et al. \[2024\]](#); the registration, ledger and negative-tested scorers follow [dinostomp \[Kwon, 2026\]](#).

3 The task and the instrument

The key. Type effectiveness and species typings come from PokeAPI’s CSV tables [Paul Hallett and PokéAPI contributors, 2026], frozen at one commit with SHA-256 hashes beside them, including the generation-specific overrides that make “Red/Blue” mean Red/Blue (Magneite is pure Electric, Ghost does nothing to Psychic). The 2,265-cell key (15 types $\times$ 151 species) is derived, never typed. Twenty-six known cells were written by hand and checked against it; six of the hand answers were wrong and the key was right.

The items. A pinned set of 400 cells, seed 0, stratified by difficulty: every cell where the Generation I and modern charts disagree (71, taken whole), then immunities (49), $4\times$ and $1/4\times$ cells (49), ordinary dual-type (132) and single-type (99) defenders (Appendix D). The commonest answer is 1 on 40.8% of items; that majority baseline is printed beside every accuracy, and it is a property of the draw: over all 2,265 cells the share is 58.8%, so a uniform draw would set a higher bar. A 100-item development set, disjoint from the pinned set, served harness debugging and was never scored against a prediction. The pinned file is committed and a test holds the sampler to it byte for byte.

The prompt. The system prompt states the rules in the game’s own words (0 “doesn’t affect”, $1/2$ “not very effective”, 2 “super effective”; a two-type defender multiplies the two factors; ignore STAB, stats and exceptions), gives the chart, and lists all 151 typings in dex order, with the instruction to use only the chart given even where it disagrees with memory. The question names the attacking type and the defender. Six options in one fixed order. Chain of thought [Wei et al., 2022b] is allowed through Inspect’s multiple-choice template [UK AI Security Institute, 2026], last line ANSWER: X; the token budget is 1,024, raised to 4,096 for reasoning models after the rule in §6 fired. The prompt is about 3,000 tokens, byte-identical for every item (Appendix B). The chart has two renderings, equal cell for cell by test: the registered default is a markdown *table* (rows attacking, columns defending); a variant registered before any pinned run renders it as *rows*, “Ground attacking: Normal 1, Fighting 1, ..., Electric 2, ...”, where every cell is named by both types and there is no axis to transpose; the rows prompt is about 6% longer (2,891 against 2,728 tokens on Gemma’s tokenizer), so its gain runs against any length effect.

Scoring and registration. Exact match is primary. Beside it, never blended: *bucket*, whether the game would print the same word ($1/4$ and $1/2$ are both “not very effective”), and *steps*, the distance on the $\log_2$ scale, with immunity three steps from everything and no answer four; those two constants are choices, so steps is reported beside the category shares in Table 3, never alone. The share of responses with no parseable answer line is its own number. Intervals are 95% Wilson [Wilson, 1927] over *items* ($n = 400$) on the three-epoch mean, an approximation for a non-binary proportion, never items times epochs; the three-epoch range is the model-side noise band. Every run saw the same items, so differences between runs are paired: a bootstrap over items on the per-item epoch means and McNemar’s exact test on per-item majority hits (Appendix E). Before any model call, mock models answering the key, the key+1, and nothing scored 1.0, 0.0 and 0.0 with parse failure 1.0, and a planted wrong chart cell moved exactly the key cells that touch it. Predictions were written and hashed before any model call; amended once, declared and re-hashed, when the budget changed the model list (Appendix F); the replication and every post-review run were registered with their own predictions before running. Registration files are never edited after their hashes are recorded; a mistake in one is a ledger entry.

4 Setup

Nine open models (Table 1): two dense families, Llama 3 [Grattafiori et al., 2024] at 1B, 3B and 8B and Gemma 3 [Gemma Team, 2025] at 4B, 12B and 27B, plus Qwen 3 [Yang et al., 2025] for the mixture-of-experts question (30B with 3.3B active; 32B dense; 235B with 22B active), and gpt-5-nano [OpenAI, 2025] as a ceiling whose size is undisclosed and which sits off the curve. Each model is pinned to one upstream host through OpenRouter with fallbacks disabled, and the quantisation the host reports is recorded: bf16 throughout Gemma, fp8 for two Qwen models and one Llama where no bf16 host existed at a usable price. qwen3-32b and gpt-5-nano reason before answering on every call; the two Qwen mixtures are the -2507 instruct variants and do not.

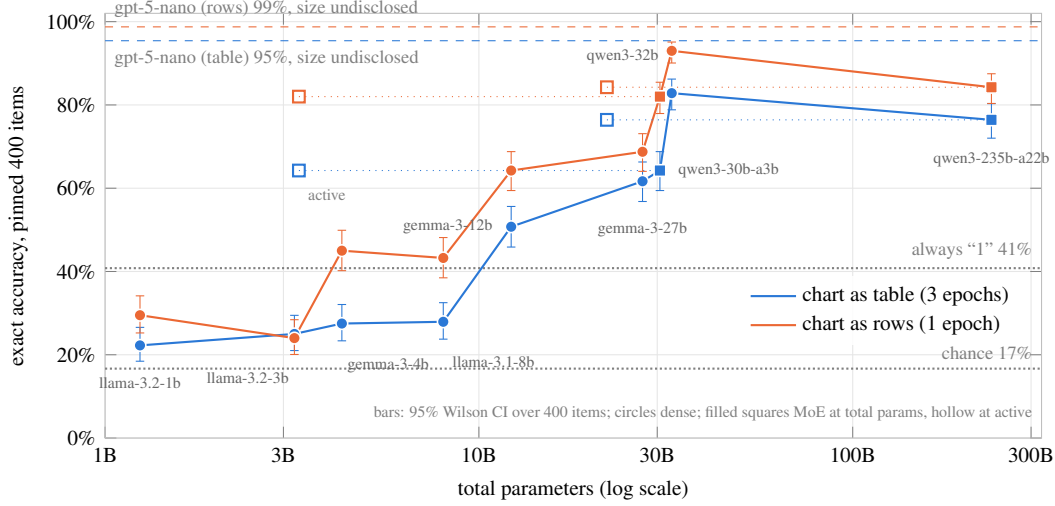


Figure 1: **Below 8B the models do not read the grid; the cleanest step is within Gemma.** Exact accuracy on the pinned 400 against total parameters (log scale). Blue: the chart as a markdown table, three epochs; orange: the same chart as one line per attacking type, one epoch. Bars are 95% Wilson intervals over 400 items. Filled squares are MoE models at their total parameter count; the hollow square joined to each by a dotted line is the same run placed at its active count. Dashed: the ceiling model, size undisclosed. Dotted: the majority baseline (always “1”, 0.41) and chance (0.17).

Table 1: **The ladder and its results on the pinned 400.** Table format: three epochs, Wilson interval over items, three-epoch range. Rows: one epoch. *quad* and *immune*: the $4 \times 1/4 \times$ and $0 \times$ strata under table ($n = 49$ each). Quantisation is what the pinned host reports (? = not reported). The qwen3-32b table value is the 4,096-token rerun; its rows value is at 1,024 tokens (2% truncated). Full model ids in the repository.

model	total (active) B	quant	host	table	95% CI	range	rows	quad	immune
llama-3.2-1b	1.2	?	Cloudflare	0.223	[0.18, 0.27]	0.035	0.295	0.03	0.10
llama-3.2-3b	3.2	bf16	Parasail	0.250	[0.21, 0.29]	0.020	0.240	0.03	0.20
gemma-3-4b	4.3	bf16	DeepInfra	0.275	[0.23, 0.32]	0.050	0.450	0.01	0.35
llama-3.1-8b	8.0	fp8	DeepInfra	0.279	[0.24, 0.33]	0.010	0.432	0.08	0.29
gemma-3-12b	12.2	bf16	DeepInfra	0.507	[0.46, 0.56]	0.020	0.642	0.33	0.74
gemma-3-27b	27.4	bf16	Novita	0.617	[0.57, 0.66]	0.022	0.688	0.21	0.80
qwen3-30b-a3b	30.5 (3.3)	bf16	CoreWeave	0.642	[0.59, 0.69]	0.050	0.820	0.21	0.78
qwen3-32b	32.8	fp8	DeepInfra	0.828	[0.79, 0.86]	0.025	0.930	0.65	0.97
qwen3-235b-a22b	235 (22)	fp8	GMICloud	0.764	[0.72, 0.80]	0.018	0.843	0.82	0.97
gpt-5-nano	undisclosed	?	Azure	0.954	[0.93, 0.97]	0.022	0.988	0.96	0.99
majority baseline				0.408			0.408		

Unparsed answers under table are 0.198 at 1B (not truncation, §6), at most 0.02 at 3B and 8B, and at most four calls in 1,200 elsewhere after the rerun in §6. Table format runs at three epochs for every model at the host’s default temperature, which OpenRouter does not expose. Rows at one epoch, then three on four rungs after review. Replication: the four rungs around the knee (8B, 12B, 27B, 30B-A3B) on a second, disjoint 400-item set (the development 100, never shown to these four, plus 300 fresh cells), table, three epochs. The unused pool held no immunity cells, so that set’s majority baseline is 0.478 rather than 0.408; the registration says so and allows 0.07 for the shift.

5 Results

Below 8B the models are not reading the grid; the cleanest step is Gemma 4B to 12B. Table 1 and Figure 1. With the chart as a grid, the four rungs from 1B to 8B score 0.22 to 0.28 against a

0.408 majority baseline, and their $4\times$ cells sit at 0.01 to 0.08: they are not failing to multiply, they are failing to read. The confound-free step is inside one family: `gemma-3-4b` and `gemma-3-12b` share a host, a quantisation and a release, and go from 0.275 to 0.507, immunities 0.35 to 0.74, quads 0.01 to 0.33; paired over items the step is $+0.233$ [$+0.185, +0.279$]. The cross-family order agrees but is also a family switch, and Gemma at 4B and Llama at 8B are not distinguishable (0.275 against 0.279; paired $+0.004$ [$-0.034, +0.042$], $p = 0.13$). The Llama steps are small: 1B to 8B is $+0.057$ [$+0.019, +0.092$] paired, and 3B to 8B is not distinguishable ($p = 0.92$).

Parameter counts did not place the mixtures; thinking did. `qwen3-30b-a3b`, with 3.3B active parameters, scores 0.642 under table and 0.820 under rows: far from the 3B and 4B rungs (0.25, 0.28), level with `gemma-3-27b` (0.617; paired $+0.026$ [$-0.014, +0.068$], $p = 0.84$, on different hosts, and the one host effect measured in this paper is 0.04), and 0.19 below `qwen3-32b` (0.828; paired $+0.186$ [$+0.146, +0.224$]). That last comparison is the only one inside a family, and it is not clean: the 32B reasons before answering on every call (618 output tokens on average) and the 30B-A3B, an instruct variant with different post-training, does not (167), so the 0.19 is sparse-versus-dense and thinking-versus-not together; a run with reasoning disabled was ignored by the host, but the thinking 8B and 14B dense rungs (§5) separate them: thinking, not density. What it does say: a sparse 30B with 3.3B active behaves like a dense model in the high twenties, not like the 3B its per-token compute matches. On the second draw it lands at 0.687, overlapping `gemma-3-27b` (0.649); we had registered the opposite, and it failed both times. The other MoE, `qwen3-235b-a22b` (22B active, also an instruct variant with no thinking), scores 0.764 under table and 0.843 under rows, below `qwen3-32b` (0.828, 0.930; paired $+0.064$ [$+0.026, +0.102$]) on this host and on a second (0.725, §5), with the same confound. Whatever limits the small dense models is not compute per token, which is what an effective-parameter view of routed models predicts [Clark et al., 2022]; on a different bounded task, Chauhan and Pendyala [2026] found the threshold in active parameters, so which count matters is task-dependent.

The rendering of the reference is a first-order effect. Rows instead of a table raises exact accuracy for nine of ten models; paired over items the gain excludes zero for all but `llama-3.2-3b`, which is at chance either way: $+0.07$ to $+0.18$ for the eight that moved, largest at 4B, 8B and the 30B mixture, $+0.03$ at the ceiling; at three epochs the gain clears the sum of both epoch ranges by 1.5 to 3 times (§5). Under rows the 4B and 8B rungs reach the majority baseline (0.45, 0.43; both intervals include 0.41) and start reading cells rather than guessing better: quads go from 0.01 to 0.22 at 4B and 0.08 to 0.14 at 8B, immunities from 0.35 to 0.49 and 0.29 to 0.55. On point estimates the knee moves to about 4B, and `qwen3-32b` (0.930) passes the 235B MoE (0.843). On the development set, 14 of Claude Haiku 4.5’s 16 table misses were the mirror cell [Anthropic, 2025].

Mirrored answers do not fall with the format below 32B, and on a rows prompt there is no axis to confuse, so those are the stored relation beating the line in front of the model. The prior leaks at the top, not the bottom. Within answer class (the contradicted cells have their own answer mix), the contradicted 0 cells score 0.21 against 0.80 at 27B and 0.55 against 0.97 at 32B; below 12B the within-class gaps are small and inconsistent in sign. We had predicted the reverse. The relabelled chart below is the clean version of this measurement.

Thinking beats size. Registered after a sixth review: the Qwen3 dense 8B and 14B, hybrid models run in the 32B’s regime (reasoning on, 4,096 tokens), table, three epochs for the 8B and two for the 14B, whose third was cut short when the credit ran out (Appendix E). The 8B scores 0.850 [0.81, 0.88] and the 14B 0.895 [0.85, 0.93], against 0.828 for the 32B: the thinking 8B beats every non-thinking model on the ladder including the 235B mixture, and Llama’s 8B by $+0.571$ [$+0.530, +0.611$] against a range sum of 0.033. Three registered predictions failed in the direction that matters: the dense ladder is not monotone (Y1) and the 30B-A3B mixture lands -0.253 [$-0.297, -0.208$] below it (Y2); the 8B clears the majority baseline by a wide margin (Y4 failed). So the knee this paper reports is a property of models that answer in one pass; with a reasoning trace the task is done at 8B, and the 0.19 gap between the 30B-A3B and the 32B is thinking, not density.

Reading, measured two ways. Two controls separate reading from memory (Table 2; both registered after review). *Recall*: the chart and the typing list removed from the prompt. Recall is 0.14 to 0.28 for every rung through 27B and the 30B mixture, below the majority baseline; about 0.51 for `qwen3-32b` and the 235B mixture; 0.665 for the ceiling. The reference raises every open model

Table 2: **Reading measured two ways, pinned 400.** *recall*: no chart and no typing list in the prompt, three epochs. *also by recall*: of the table run’s hits among the 329 items where the Generation I and modern charts agree, the share the recall run also gets right. *relabelled*: the chart, the typing list and the question under a seeded permutation of the 15 type names, one epoch; *differs*: accuracy on the 71 contradicted cells, real chart $\rightarrow$ relabelled. qwen3-32b ran recall and relabelled at 1,024 tokens (11% and 20% truncated), so both are floors.

model	recall	95% CI	table	gain	also by recall	relabelled	differs
llama-3.2-1b	0.142	[0.11, 0.18]	0.223	+0.081	0.07	0.235	0.22 $\rightarrow$ 0.23
llama-3.2-3b	0.204	[0.17, 0.25]	0.250	+0.046	0.49	0.193	0.33 $\rightarrow$ 0.32
gemma-3-4b	0.172	[0.14, 0.21]	0.275	+0.103	0.45	0.278	0.39 $\rightarrow$ 0.30
llama-3.1-8b	0.190	[0.15, 0.23]	0.279	+0.089	0.24	0.233	0.31 $\rightarrow$ 0.20
gemma-3-12b	0.266	[0.22, 0.31]	0.507	+0.241	0.37	0.517	0.43 $\rightarrow$ 0.52
gemma-3-27b	0.279	[0.24, 0.33]	0.617	+0.338	0.28	0.650	0.46 $\rightarrow$ 0.63
qwen3-30b-a3b	0.248	[0.21, 0.29]	0.642	+0.394	0.25	0.700	0.44 $\rightarrow$ 0.65
qwen3-32b	0.507	[0.46, 0.56]	0.828	+0.321	0.63	0.693	0.69 $\rightarrow$ 0.66
qwen3-235b-a22b	0.510	[0.46, 0.56]	0.764	+0.254	0.65	0.770	0.60 $\rightarrow$ 0.82
gpt-5-nano	0.665	[0.62, 0.71]	0.954	+0.289	0.83	0.943	0.87 $\rightarrow$ 0.94
majority baseline	0.408		0.408			0.408	

by more than the sum of the two epoch ranges except llama-3.2-3b (W1, 8 of 9). On the 329 items where the given chart and the memorised one agree, the share of a model’s table hits that recall also gets is 0.07 to 0.49 below 32B and 0.63 to 0.83 from 32B up; we had predicted it would pass 0.5 from 27B (W2); gemma-3-27b and qwen3-30b-a3b read well and remember almost nothing. *Relabelled*: the same chart, list and question under a seeded permutation of the fifteen type names, so the key is unchanged and memory of the real chart is useless. At a matched token budget every model lands within 0.06 of its real-chart score (X1), and the three models with a memory share above 0.6 were predicted to fall by more than 0.05 (X2) and did not: the 235B mixture and the ceiling moved under 0.02, and the 32B’s drop is its 1,024-token budget (0.697 at that budget with the real chart). Where the memory does act is the 71 contradicted cells: with the names permuted, accuracy on them rises at every rung from 12B up, by 0.07 to 0.22 (Table 2). On the 329 agreeing cells the same run costs the 235B mixture 0.040 and the ceiling 0.029 and moves the 27B and the 30B mixture by 0.000 and +0.024: the prior is a tax of 0.07 to 0.22 on the cells it contradicts and a subsidy of at most 0.04 on the rest. The ladder measures reading at every rung above chance.

The word is not much easier than the number; the lean is toward “something happens”. Bucket accuracy exceeds exact by 0.00 to 0.07 across the twenty runs, about 0.02 above 12B: misses are mostly not the adjacent multiplier. Mean $\log_2$ steps off (Table 3) falls from 1.94 at 1B to 0.08 at the ceiling, and rows shortens it at every rung but 3B. Both registered clauses failed for most runs. With options in a fixed order, the share of predicted “1” sits near its 40.8% target from 12B up, except the 30B mixture (52%). At 3B, 4B and 8B it does not, and not toward the majority: gemma-3-4b answers “1” on 10% of items and “2” on 68%, llama-3.2-3b 11% and 56%. With the options shuffled per item they still answer “2” on 0.63 and 0.48 of items: a value lean, not a letter lean.

Error analysis: what a miss is, by size. Every miss under table format is filed into one of five categories, first match wins (Table 3; one verbatim sample per category in Appendix C): *unparsed*; *mirrored* (the key with attacker and defender swapped, where that differs; under table this cannot separate a transposed read from the memorised direction, and only the rows rendering can); *immunity* (a 0 answered as something else, or the reverse); *one step, same word* (2 for 4, $\frac{1}{4}$ for $\frac{1}{2}$, or the reverse); *lookup* (everything else). The categories are outcomes, not processes: a one-step miss can be two cells read right and multiplied wrong or one cell misread, so the taxonomy bounds the multiply failure from above. **Multiplying is at most a small part of the failure:** the one-step share is 0.03 to 0.13 at every rung. **Immunity misses change kind with size, not count:** filed after mirrored, they are 0.11 to 0.20 of misses at 1B to 8B and 0.05 or less from 27B up; filed first, the share is flat, 0.13 to 0.24 at every rung but 235B (0.05), because from 27B up most immunity-type misses are also the mirror cell (67 of 74 at 27B, 27 of 154 at 8B). Small models get immunities wrong in every direction; large ones get them wrong by reading the grid backwards. **Mirrored answers grow with size and then shrink:** 0.16 to 0.23 of misses at 1B to 8B, 0.32 at 12B, 0.48 to

Table 3: **What a miss is, by model, table format, pinned 400×3 epochs.** *steps* is the mean $\log_2$ distance between answer and key over all items (immunity counts as three, no answer as four). The remaining columns are shares of that model’s misses; categories are assigned in the order listed in the text, first match wins.

model	misses	of 1,200	steps	unparsed	mirrored	immunity	one step	lookup
llama-3.2-1b	933	0.78	1.94	0.26	0.16	0.20	0.03	0.35
llama-3.2-3b	900	0.75	1.36	0.02	0.20	0.17	0.10	0.52
gemma-3-4b	870	0.72	1.23	0.00	0.23	0.11	0.10	0.57
llama-3.1-8b	865	0.72	1.30	0.03	0.22	0.15	0.07	0.54
gemma-3-12b	591	0.49	0.81	0.00	0.32	0.09	0.06	0.53
gemma-3-27b	460	0.38	0.67	0.00	0.60	0.02	0.07	0.32
qwen3-30b-a3b	429	0.36	0.65	0.00	0.48	0.09	0.04	0.40
qwen3-32b	206	0.17	0.28	0.00	0.38	0.05	0.11	0.45
qwen3-235b-a22b	283	0.24	0.33	0.00	0.53	0.02	0.10	0.35
gpt-5-nano	55	0.05	0.08	0.07	0.36	0.00	0.13	0.44

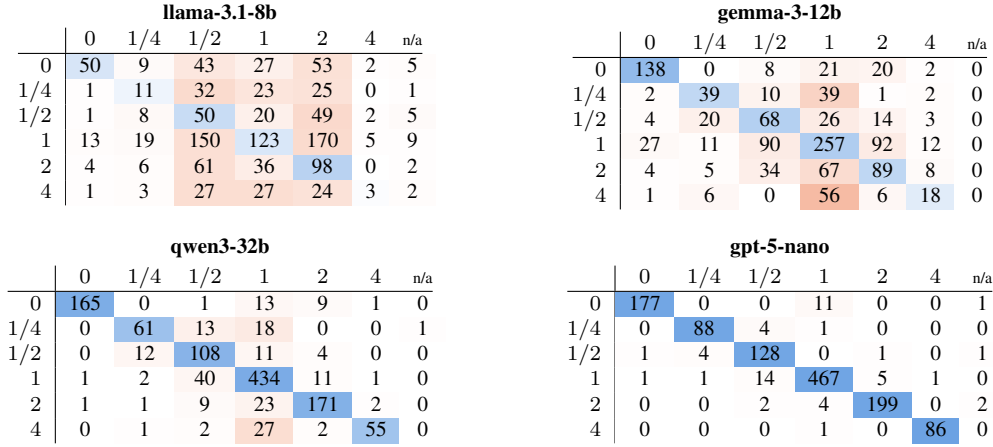


Figure 2: **Confusion matrices, table format, three rungs and the ceiling model.** Rows are the target multiplier, columns the prediction (last column: no answer parsed); counts over 1,200 scored samples. Shading is row-normalised: blue on the diagonal, orange off it, darker is a larger share of that row.

0.60 at 27B to 235B, then 0.38 at 32B and 0.36 at the ceiling: a model that can read the grid misses by reading it backwards or letting the memorised direction win. The confusion matrices (Figure 2) show the same progression.

Inside the lookup residual. *lookup* is the largest miss category at every rung. A second pass reads each lookup miss for the typing the model states for the defender and compares it with the truth. At 1B, 45% of lookup misses state a wrong typing and 30% none: it is not finding the line. At 3B and 4B the wrong-typing share is 15 to 16%; at 8B and 12B, 4 to 8%; from 27B up, 0 to 1%: the rest state the right typing or none, and miss in the chart.

Second draw. The second 400 is a robustness check on a second draw, not a replication of the same distribution. On it (Table 5, Appendix E), the order $8B < 12B < 27B$ holds with steps of 0.15 to 0.17 against epoch ranges of 0.005 to 0.045; the MoE’s interval overlaps 27B’s and not 8B’s; no model moved more than 0.05 between two disjoint draws. One registered clause failed: gemma-3-12b ties this set’s majority baseline (0.480 against 0.478).

Runs made after review. Three external reviews asked for seven further runs, each registered with its predictions before running (Appendix E). The recall and relabelled passes are above; the 32B without thinking was discarded (§5). (i) Rows at three epochs on the four knee rungs: the gain over table exceeds the sum of both epoch ranges (V1) and the one-epoch number lies within 0.03 of

the three-epoch mean (V2), both held for all four (Table 6). (ii) `qwen3-235b-a22b` on a second host, DeepInfra instead of GMICloud, both fp8: 0.725 against 0.764, inside the first host’s interval (V3), at the edge (host effect 0.04). (iii) Options shuffled per item for the two models that lean on “2”: the value share moved from 0.68 to 0.63 and from 0.56 to 0.48 (W4 value clause held). Its letter clause could not be scored (D-014). (iv) `gemma-3-12b` at temperature 0: 0.502, range 0.015, inside the default-temperature interval (W5).

6 What the harness caught

The ledger has seventeen entries against the harness or the plan and thirteen observations (Appendix G). The two that changed numbers, and the one that would have, are here.

A token budget read as capability. `gpt-5-nano` at 1,024 tokens returned an empty completion on 30% of development items: hidden reasoning consumed the budget before the answer line was written. The parse-failure count flagged it as truncation; at 4,096 the model scored 0.98. `qwen3-32b`, a thinking model, truncated 14% of its table calls at 1,024 and scored 0.697; at 4,096 it scored 0.828 with one truncation in 1,200. A 0.13 swing from the budget alone. The 1B leaves 19.8% of its table answers unparsed at 1,024; rerun at 4,096 it scores 0.211 with 19.3% unparsed and no call at the cap: missing answer lines, not truncation (D-017).

The provider behind the provider. One `llama-3.2-3b` development run came back at 0.17 with 31% of completions cut mid-word: the aggregator had split the calls across two hosts and one returned HTTP 500 on 31 of its 70, scored as wrong answers. Pinned to the healthy host, the model scored 0.18, chance, the real number. A census then found `qwen3-235b` served by ten hosts of unknown quantisation in one run; every pinned run here uses one host, recorded per call.

A metric that lied at three epochs. The in-log parse-failure metric read 0.94 on a run whose per-sample rate was 0.198: epoch reduction drops the string field the metric counted. The analysis recomputes every rate from the unreduced samples; the metric became a numeric scorer.

7 Limitations

One task and one domain. The reference is partly memorised; the relabelled run is the one-epoch, table-only answer to that, and its 32B point is a 1,024-token floor. Neither item set is a uniform draw of the 2,265 cells (the pinned set takes every contradicted cell; the second draw has no immunities), so absolute accuracies are not comparable across them; a uniform third set is the fix. Item sets are 400, so stratum-level intervals ($n = 49$) are wide. Hosts, quantisations, thinking mode and post-training differ across families and are reported, not controlled, and the hosts serve bf16 or fp8 where on-device deployments are usually 4-bit; the only step free of all of them is Gemma 4B to 12B, and the cross-family order is consistent with it, not a threshold in its own right. Temperature is each host’s default (one rung at 0: -0.005). The second draw covered four rungs, table only. The plan was cut to a \$7 budget before any pinned run (Appendix F).

8 Conclusion

The exact accuracy of a model on this task is “how often it finds one line in a 151-line list, reads two cells of a 225-cell chart, and multiplies”, with everything it needs in front of it. Its deployment reading: a task of this shape is not something the 1B to 8B class does from a grid in one pass, is something the 4B class begins to do from lines, and reaches 0.83 to 0.93 at 32B in an open model. The ladder said four things a single number would not: above chance the score is reading, since recall alone sits under the majority baseline through 27B and relabelling the types moves no model by more than 0.06 at a matched budget; family matters as much as size below 12B; a reasoning trace is worth more than any rung, an 8B with one beating a 235B without; and how the reference is rendered is worth up to a rung. Nine of its thirty registered predictions failed outright. Whether this transfers to a reference no model has seen is an inference, not a measurement; the harness, key, item sets, registration files and ledger are released so that it can be made one.

LLM usage statement

Claude Fable 5 was used agenticly, with tool access to the repository, as a coding assistant: it wrote most of the harness, the analysis, the registration files, the ledger entries and the draft prose. The author set the direction, chose the task, the models and every stopping point, read the misses, and is responsible for every claim. No language model scored any result: every number is a count over logged answers against a code-generated key.

References

- Anthropic. Claude Haiku 4.5. <https://www.anthropic.com/claude/haiku>, released 2025-10-15, 2025. Parameter count not disclosed.
- Stella Biderman et al. Lessons from the trenches on reproducible evaluation of language models. <https://arxiv.org/abs/2405.14782>, 2024.
- Aman Chauhan and Vishnu Pendyala. Benchmarking LLM-guided control-plane policies for back-end fault isolation in HAProxy. <https://arxiv.org/abs/2608.10532>, 2026.
- Aidan Clark, Diego de las Casas, Aurelia Guy, et al. Unified scaling laws for routed language models. <https://arxiv.org/abs/2202.01169>, 2022.
- William Fedus, Barret Zoph, and Noam Shazeer. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research*, 2022. URL <https://arxiv.org/abs/2101.03961>. arXiv:2101.03961.
- Gemma Team. Gemma 3 technical report. <https://arxiv.org/abs/2503.19786>, 2025.
- Aaron Grattafiori et al. The Llama 3 herd of models. <https://arxiv.org/abs/2407.21783>, 2024.
- Tom Gunter, Zirui Wang, Chong Wang, et al. Apple intelligence foundation language models. <https://arxiv.org/abs/2407.21075>, 2024.
- Cheng-Ping Hsieh, Simeng Sun, Samuel Kriman, et al. RULER: What’s the real context size of your long-context language models? In *Conference on Language Modeling (COLM)*, 2024. URL <https://arxiv.org/abs/2404.06654>. arXiv:2404.06654.
- Jared Kaplan, Sam McCandlish, Tom Henighan, et al. Scaling laws for neural language models. <https://arxiv.org/abs/2001.08361>, 2020.
- Alex Kwon. dinostomp: negative-tested checks for dataset and evaluation validity. <https://github.com/collapseindex/dinostomp>, 2026.
- Zechun Liu, Changsheng Zhao, Forrest Iandola, Chen Lai, Yuandong Tian, Igor Fedorov, Yunyang Xiong, Ernie Chang, Yangyang Shi, Raghuraman Krishnamoorthi, Liangzhen Lai, and Vikas Chandra. MobileLLM: Optimizing sub-billion parameter language models for on-device use cases. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *PMLR*, pages 32431–32454, 2024. URL <https://proceedings.mlr.press/v235/liu24ce.html>.
- OpenAI. GPT-5 nano, model documentation. <https://developers.openai.com/api/docs/models/gpt-5-nano>, snapshot gpt-5-nano-2025-08-07, 2025. Parameter count not disclosed.
- Paul Hallett and PokéAPI contributors. PokeAPI: the RESTful Pokémon API, data tables. <https://github.com/PokeAPI/pokeapi>, commit 7af36d9, 2026. BSD-3-Clause. Pokémon and Pokémon character names are trademarks of Nintendo.
- Pouya Pezeshkpour and Estevam Hruschka. Large language models sensitivity to The order of options in multiple-choice questions. In *Findings of the Association for Computational Linguistics: NAACL 2024*, 2024. URL <https://arxiv.org/abs/2308.11483>. arXiv:2308.11483.
- Rylan Schaeffer, Brando Miranda, and Sanmi Koyejo. Are emergent abilities of large language models a mirage? <https://arxiv.org/abs/2304.15004>, 2023.
- Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. Quantifying language models’ sensitivity to spurious features in prompt design or: How I learned to start worrying about prompt formatting. In *International Conference on Learning Representations (ICLR)*, 2024. URL <https://arxiv.org/abs/2310.11324>. arXiv:2310.11324.
- Yuan Sui, Mengyu Zhou, Mingjie Zhou, et al. Table meets LLM: Can large language models understand structured table data? a benchmark and empirical study. In *Proceedings of the*

- 17th ACM International Conference on Web Search and Data Mining (WSDM), 2024. URL <https://arxiv.org/abs/2305.13062>. arXiv:2305.13062.
- UK AI Security Institute. Inspect: a framework for large language model evaluations. <https://inspect.aisi.org.uk>, version 0.3.260, 2026.
- Jason Wei, Yi Tay, Rishi Bommasani, et al. Emergent abilities of large language models. *Transactions on Machine Learning Research*, 2022a. URL <https://arxiv.org/abs/2206.07682>. arXiv:2206.07682.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, et al. Chain-of-thought prompting elicits reasoning in large language models. <https://arxiv.org/abs/2201.11903>, 2022b.
- Edwin B. Wilson. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927. URL <https://doi.org/10.1080/01621459.1927.10502953>.
- Zijie Xu, Wenjun Ke, Peng Wang, Guozheng Li, Qingjian Ni, Jiajun Liu, Ziyu Shang, and Jing Zhou. Benchmarking and enhancing rule knowledge-driven reasoning of large language models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2026. URL <https://ojs.aaai.org/index.php/AAAI/article/view/40714>.
- An Yang et al. Qwen3 technical report. <https://arxiv.org/abs/2505.09388>, 2025.
- Chujie Zheng, Hao Zhou, Fandong Meng, Jie Zhou, and Minlie Huang. Large language models are not robust multiple choice selectors. In *International Conference on Learning Representations (ICLR)*, 2024. URL <https://arxiv.org/abs/2309.03882>. arXiv:2309.03882.

A Registered predictions and status

The nine that failed outright:

registered prediction	what happened
The game’s word is at least three points easier than the number.	Gaps of 0.00 to 0.07; misses are mostly not the adjacent multiplier.
The contradicted cells hurt the smallest model most.	Reversed: the gap appears from 12B up and is absent below.
No model leans on one option letter.	Below 12B they lean hard, toward “super effective”, not the majority.
Rows format halves the mirrored-cell misses.	Only for the two strongest models; below 32B those misses are the prior, not the grid.
The 30B MoE lands with the 3B to 4B rungs.	It lands with the 27B and 32B dense models, on both item sets.
The models with a high memory share drop when the chart is relabelled.	They do not: the 235B and the ceiling move under 0.02 overall; on the agreeing cells they lose 0.04 and 0.03, the subsidy X2 predicted, under its threshold.
The Qwen3 dense ladder 8B < 14B < 32B is monotone.	The 32B is below both (0.828 against 0.850 and 0.896).
The 30B MoE scores at or above the thinking 14B dense model.	0.25 below it.
The thinking Qwen3 8B sits at or below the majority baseline.	0.850: a thinking 8B beats every one-pass model on the ladder.

HELD: the prediction as registered is supported by the measurement. FAILED: it is not, and the row says how. NOT RUN: registered in the original plan and retired by the addendum before any measurement; the row exists so a reader cannot infer it.

Table 4: **Predictions and status.** Registered in seven hashed files: the plan (P), the addendum (A, after the budget changed the model list), the replication (R), and the runs after the first (V), second (W), third (X) and sixth (Y) external reads; each hash is recorded in the repository README before the corresponding run.

Prediction	Evidence	Status
P1: the three-epoch range of exact accuracy is between 1 and 5 points for every model.	Tab. 1	HELD, 10 of 10 (0.010 to 0.050)
P2: giving the typing helps more than giving the chart (recall conditions).	n/a	NOT RUN
P3: the best model reaches 0.95 exact.	Tab. 1	HELD: the ceiling model, 0.954; no open model under table
P4: models follow a provided chart that contradicts the memorised one.	n/a	NOT RUN as a condition; the <i>differs</i> stratum under the Gen I chart is the nearest measurement (§5)
P5, P6: recall-condition strata and the Ghost→Psychic row.	n/a	NOT RUN
A1: bucket accuracy exceeds exact by at least 3 points; half of misses are one step off.	§5	FAILED: the bucket clause on 11 of 20 runs (gaps of 0.00 to 0.07), the one-step clause on all 20
A2: exact accuracy on 4× cells is below single-type cells by more than the range.	Tab. 1	HELD, 9 of 10; the bucket clause of A2 FAILED everywhere
A3: wrong-Pokémon lookups explain under 5% of items.	§5	HELD on the model it was hand-scored for (Haiku, 0 of 100); not hand-scored for the ladder
A4: the contradicted cells fall below ordinary cells for the smallest model and not the largest.	§5	FAILED, reversed: the gap is 0.09 to 0.28 from 12B up and absent below 12B

continued on next page

Table 4, continued

Prediction	Evidence	Status
A5: the predicted share of “1” is within 10 points of its target share for every model.	§5	FAILED for 6 of 20 runs: five below 12B, and the 30B mixture under table (52%)
A6: rows beats table by more than the range for every model.	Tab. 1	HELD, 9 of 10
A6: mirrored-cell misses fall by at least half under rows.	§5	FAILED, 8 of 10: below 32B the mirrored answers are the prior, not the grid
A7: accuracy is monotone in size within each dense family, each step beyond the ranges.	Tab. 1	HELD for Gemma; Llama is monotone but its steps are inside the ranges
A7: the 30B MoE lands nearer the 3B to 4B rungs than the 32B dense model.	§5	FAILED, twice
A7: the smallest model to reach 0.90 under table is at or above 12B.	Tab. 1	NOT SHOWN: no open model reached 0.90 under table; under rows the smallest is 32B
R1: on the replication set, 8B’s interval is below the majority and 12B’s above it.	Tab. 5	half HELD: 8B below; 12B ties (0.480 against 0.478)
R2: $8B < 12B < 27B$ with each step beyond both ranges.	Tab. 5	HELD
R3: the MoE’s interval overlaps 27B’s and not 8B’s.	Tab. 5	HELD
R4: each model inside its pinned interval or within the 0.07 baseline allowance.	Tab. 5	HELD, 4 of 4
The knee is a property of the model alone.	§5	NOT CLAIMED: it moved from 8B to 12B to about 4B with the rendering of the reference
V1: rows minus table at three epochs exceeds the sum of both epoch ranges, four rungs.	Tab. 6	HELD, 4 of 4
V2: the one-epoch rows number lies within 0.03 of the three-epoch mean.	Tab. 6	HELD, 4 of 4
V3: the 235B MoE on a second host lands inside the first host’s interval.	App. E	HELD, 0.725 in [0.720, 0.803]
W1: the reference raises every open model by more than the sum of both epoch ranges.	Tab. 2	HELD, 8 of 9; llama-3.2-3b misses by 0.002, at chance either way
W2: the share of table hits also hit by recall is below 0.5 at 8B and under, and above 0.5 from 27B up.	Tab. 2	half HELD: under 0.5 at 8B and below, over 0.5 only from 32B; 27B and the 30B MoE at 0.28 and 0.25
W3: recall is below the majority baseline at 8B and under.	Tab. 2	HELD, and through 27B
W4: under shuffled options the value share of “2” stays within 10 points, and no letter exceeds 0.40.	§5	value clause HELD; letter clause
W5: temperature 0 lands inside the default-temperature interval with no larger range.	§5	NOT MEASURABLE (D-014)
X1: under the relabelled chart every open model at 27B and under, and the 30B MoE, stays within 0.05 of its table mean.	Tab. 2	HELD, 0.502, range 0.015
X2: the 32B, the 235B MoE and the ceiling each fall by more than 0.05 under the relabelled chart.	Tab. 2	HELD, 5 of 7: the 3B fell 0.057 (at chance either way), the 30B MoE rose 0.057
X3: $8B < 12B < 27B$ under the relabelled chart, each step beyond 0.030.	Tab. 2	FAILED: 235B and the ceiling moved under 0.02; the 32B’s 0.136 drop is its 1,024-token budget (0.697 at that budget with the real chart)
Y1: Qwen3 dense $8B < 14B < 32B$, each step beyond the paired interval.	§5	HELD:
Y2: the 30B-A3B mixture scores at or above the thinking 14B dense model.	§5	0.233 < 0.517 < 0.650
Y3: the thinking Qwen3 8B exceeds Llama 8B by more than the sum of both ranges.	§5	FAILED: 8B → 14B +0.045 [+0.010, +0.081], 14B → 32B −0.067 [−0.097, −0.037]
Y4: Qwen3 8B at or below the majority baseline, 14B above it.	§5	FAILED: 0.642 against 0.895
X4, X5: the 32B with reasoning disabled still beats the 30B MoE (density) and falls below its thinking score (thinking).	App. E	HELD, +0.571 [+0.530, +0.611] against a range sum of 0.033
		FAILED: both far above, 0.850 and 0.895
		NOT SCORABLE: the host ignored the request; reasoning on 1,200 of 1,200 calls; run discarded as the registration requires (D-016)

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Table 4, continued

Prediction	Evidence	Status
Anything about what these models know about Pokémon beyond the chart.	n/a	NOT CLAIMED; the recall pass measures the chart only

B The prompt, as sent

System prompt (rules, then the chart, then the list; the chart and list are cut here and are complete in the prompt):

You are answering questions about type matchups in Pokemon Red and Blue (Generation I, the original 151 Pokemon, as typed in those games).

Answer with the damage multiplier from type effectiveness alone: 0 (the move doesn't affect the defender), 1/2 (not very effective), 1 (normal damage), or 2 (super effective). A defender with two types multiplies the two factors, so 1/4 and 4 are possible.

Ignore STAB, stats, move power, and any move-specific exceptions. Only the attacking type against the defender's type or types counts.

Use only the type chart below (the Generation I chart), even where it disagrees with what you remember about the games. Rows are the attacking type, columns the defending type.

```
| attack \{} defend | normal | fighting | flying | poison
| ground | rock | bug | ghost | fire | water | grass |
electric | psychic | ice | dragon |
|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
| normal | 1 | 1 | 1 | 1 | 1 | 1/2 | 1 | 0 | 1 | 1 | 1 | 1 |
1 | 1 | 1 |
[... 13 more rows]
```

The typing of every Pokemon, as in Red and Blue:

```
1. Bulbasaur: Grass/Poison
2. Ivysaur: Grass/Poison
3. Venusaur: Grass/Poison
[... 148 more lines]
```

User turn (Inspect's chain-of-thought multiple-choice template around the item; item i0000 of the pinned set):

Answer the following multiple choice question. The last line of your response should be of the following format: 'ANSWER: \$LETTER' (without quotes) where LETTER is one of A,B,C,D,E,F. Think step by step before answering.

A Ice-type move hits Vulpix. What is the damage multiplier from type effectiveness alone?

- A) 0
- B) 1/4
- C) 1/2
- D) 1
- E) 2
- F) 4

The rows rendering replaces the table block with fifteen lines of the form "Ground attacking: Normal 1, Fighting 1, ..."; nothing else changes.

C One sample per miss category

Verbatim model output, cut in the middle where long, from the table-format pinned runs; the category is the one §5 assigns. The first is a hit for the shape of a correct answer.

hit. Fighting vs Butterfree (Bug/Flying); key 1/4; answered 1/4; item i0001, epoch 1; gemma-3-12b-it

```
Butterfree has two types: Bug and Flying. A Fighting-type
move hits a Bug-type with a multiplier of 1/2. A
Fighting-type move hits a Flying-type with a multiplier
of 1/2. The total damage multiplier is (1/2) * (1/2) = 1/4.
ANSWER: B
```

mirrored. Ice vs Vulpix (Fire); key 1; answered 2; item i0000, epoch 1; gemma-3-27b-it

```
The attacking type is Ice, and the defending Pokemon is
Vulpix. Vulpix is a Fire-type Pokemon. Looking at the type
chart, Ice attacks against Fire-type Pokemon have a damage
multiplier of 2 (super effective). ANSWER: E
```

immunity. Bug vs Beedrill (Bug/Poison); key 2; answered 0; item i0008, epoch 1; gemma-3-4b-it

```
Let's analyze the type matchup. Beedrill's type is
Bug/Poison. We need to determine the damage multiplier
against a Bug-type move. Looking at the type chart, Bug
attacks have a multiplier of 1 against Bug. ANSWER: A
```

one step, same word. Rock vs Golem (Rock/Ground); key 1/2; answered 1/4; item i0055, epoch 1; qwen3-32b

```
The Rock-type move's effectiveness against Golem
(Rock/Ground) is calculated by checking each of Golem's
types. - **Rock vs Rock**: The multiplier is **1/2** (Rock
moves are not very effective against Rock). - **Rock vs
Ground**: The multiplier is **1/2** (Rock moves are not
very effective against Ground). Multiplying these together:
**1/2 x 1/2 = 1/4**. ANSWER: B
```

lookup. Bug vs Aerodactyl (Rock/Flying); key 1/2; answered 2; item i0002, epoch 1; llama-3.1-8b-instruct

```
To determine the damage multiplier from type effectiveness
alone, we need to look at the type chart. Aerodactyl is
a Rock/Flying-type Pokémon. Bug-type moves are super
effective against Rock-type Pokémon, so the damage
multiplier from type effectiveness alone is 2. ANSWER:
E
```

unparsed. Normal vs Chansey (Normal); key 1; answered none; item i0011, epoch 1; llama-3.2-1b-instruct

```
To answer this question, we need to look at the type chart
and find the attacking type (Chansey) and the defending
type (Normal). From the type chart, we can see that
Chansey is Normal-type and Normal is the defending type.
Now, we can calculate the damage multiplier using the
formula: (attacking type multiplier) x (defending type
multiplier) The attacking type multiplier for Normal is 1.
The defending type multiplier for Normal is 1/2. Therefore,
the damage multiplier is 1 x 1/2 = 1/2. The correct answer
is C) 1/2.
```

Table 5: **Replication on a second, disjoint 400-item set**, table format, three epochs. *shift* is replication minus pinned. The registration allowed 0.07 for this set’s higher majority baseline.

model	replication	95% CI	pinned	shift
llama-3.1-8b	0.326	[0.28, 0.37]	0.279	+0.047
gemma-3-12b	0.480	[0.43, 0.53]	0.507	−0.027
gemma-3-27b	0.649	[0.60, 0.69]	0.617	+0.032
qwen3-30b-a3b	0.687	[0.64, 0.73]	0.642	+0.044
majority baseline	0.478		0.408	

Table 6: **Rows format at three epochs, four rungs, pinned 400**. Gain is rows minus table, both three-epoch means; *ranges* is the sum of the two three-epoch ranges the gain has to clear.

model	table	rows, 3 ep.	95% CI	gain	ranges	rows, 1 ep.
llama-3.1-8b	0.279	0.412	[0.36, 0.46]	+0.133	0.055	0.432
gemma-3-12b	0.507	0.646	[0.60, 0.69]	+0.138	0.048	0.642
gemma-3-27b	0.617	0.680	[0.63, 0.72]	+0.063	0.042	0.688
qwen3-30b-a3b	0.642	0.828	[0.79, 0.86]	+0.186	0.065	0.820

D Item-set composition

stratum	0	1/4	1/2	1	2	4	total
differs	14	0	0	20	26	11	71
immune	49	0	0	0	0	0	49
quad	0	31	0	0	0	18	49
dual	0	0	33	67	32	0	132
single	0	0	12	76	11	0	99
total	63	31	45	163	69	29	400

Answer class by stratum for the pinned 400. *immune* is all-0 and *quad* is all-1/4-or-4 by definition; *differs* holds every cell where the Generation I and modern charts disagree. The replication set is the development 100 (immune 14, quad 15, dual 41, single 30) plus 300 fresh cells (quad 19, dual 191, single 90); its commonest answer is 1 on 47.8% of items.

E Replication and post-review tables

The second-host run: qwen3-235b-a22b, table, three epochs, pinned 400, DeepInfra (fp8) instead of GMICloud (fp8), fallbacks disabled. Prediction V3: the DeepInfra mean lies inside GMICloud’s interval [0.720, 0.803]; above 0.828 the ordering against qwen3-32b would flip. Result 0.725, interval [0.679, 0.766], range 0.013, no host-side errors in 1,200 calls.

The relabelled chart (REVIEW3.md run 1): `src.key.type_permutation(0)` maps normal→bug, fighting→electric, flying→grass, poison→ice, ground→normal, rock→dragon, bug→ground, ghost→rock, fire→water, water→fighting, grass→ghost, electric→fire, psychic→flying, ice→psychic, dragon→poison; a test holds the relabelled chart equal to the real one cell for cell under the inverse map. One epoch, table format, pinned 400, same hosts; qwen3-32b at 1,024 tokens. The 32B without thinking (REVIEW3.md run 2) was requested through OpenRouter’s reasoning: {enabled: false}; DeepInfra served reasoning on every one of 1,200 calls (557 output tokens on average, 12.6% truncated at 1,024), so the run is discarded and X4 and X5 are unscored. Its score, 0.686, replicates the 1,024-token thinking floor (0.697).

Qwen3 dense rungs (REVIEW4.md), table, 4,096 tokens, reasoning on; the 8B at three epochs, the 14B at two complete epochs plus 129 items of a third (825 scored calls) before the account’s credit ran out (D-018): qwen3-8b (Alibaba, quantisation not reported) 0.850 [0.81, 0.88], range 0.022,

truncation 0.000; qwen3-14b (DeepInfra fp8) 0.895 [0.85, 0.93], range 0.005, truncation 0.000. Quads 0.75 and 0.86, immunities 1.00 and 1.00.

Paired tests over the pinned items (`src.paired`): bootstrap interval on the mean per-item difference (epoch means, 10,000 resamples) and McNemar’s exact test on per-item majority hits, discordant counts b/c .

comparison (table format)	difference	paired 95%	$b/c, p$
Gemma 4B → 12B	+0.233	[+0.185, +0.279]	41/134, < 0.001
Llama 1B → 8B	+0.057	[+0.019, +0.092]	43/73, 0.007
Llama 3B → 8B	+0.029	[−0.005, +0.063]	53/55, 0.92
Gemma 4B vs Llama 8B	+0.004	[−0.034, +0.042]	65/48, 0.13
Gemma 27B vs Qwen 30B-A3B	+0.026	[−0.014, +0.068]	50/53, 0.84
Qwen 30B-A3B → 32B	+0.186	[+0.146, +0.224]	16/102, < 0.001
Qwen 235B-A22B → 32B	+0.064	[+0.026, +0.102]	31/62, 0.002
rows minus table, nine models	+0.033 to +0.177	all exclude 0	$p \leq 0.013$ except the ceiling (0.21)
rows minus table, llama-3.2-3b	−0.010	[−0.047, +0.028]	37/41, 0.73

F Reproduction

Environment. Inspect 0.3.260 [UK AI Security Institute, 2026] over OpenRouter, one pinned host per model with fallbacks disabled, provider-default temperature, token budget 1,024 or 4,096 as stated. The host pin is passed to Inspect as a model argument: `-M provider={"order":["<host>"], "allow_fallbacks":false}`. Python 3.13 in a project virtual environment; every dependency is pinned in `requirements.txt`.

Commands. From the repository root (<https://github.com/collapseindex/pokemon-gen1-eval>):

```
python -m src.key # derive the key, print the 26 known
cells
python -m src.sample -n 400 -seed 0 # reproduces the
committed pinned set byte for byte
pytest # 21 tests, no API calls
python -m src.run_ladder # table x3 for every model in
src/models.py, then rows x1
python -m src.analyze -log-dir logs/pinned # results JSON,
curve CSV, 95% intervals
python -m src.error_analysis -log-dir logs/pinned # miss
categories, confusion matrices
python writeup/paper/make_figures.py # Figures 1 and 2 and
Table 3 from those files
```

Deviation from the plan. The original plan registered four conditions (recall with no reference, typing shown inline, the full reference, and a modern chart scored against itself), three closed models, and a \$30 ceiling. Multiplied out it cost about \$157; the available budget was \$7. The plan was not edited. An addendum, hashed before any pinned run, replaced it with one condition (the full reference in context), an open-model size ladder, and a \$5 ceiling; the retired conditions stay in the code as parameters, and the recall condition was run after review. The replication and the 32B rerun, registered after the ceiling was set, took the spend at that point to \$5.72; that overrun is a ledger entry rather than an absorbed one.

Cost. Every run in this paper, including the development runs, the discarded runs of \$6, the replication, the reruns, and the eight post-review runs: \$12.05 of OpenRouter credit, read from the account balance before and after (\$7.22 through the first review round, \$1.88 for the second, \$1.35 for the third, \$0.12 for the 1B rerun, \$1.48 for the Qwen3 dense rungs, whose reasoning traces cost about twice the estimate). The nine-model table pass was about \$2.30; the 32B rerun at 4,096 tokens about \$0.80; the four-model rows pass at three epochs about \$1.05; the second-host run about \$0.45; the recall pass, the shuffle and the temperature run \$1.88 together against \$1.32 estimated; the relabelled pass and the discarded no-thinking run \$1.35 against \$1.15.

Frozen inputs. SHA-256 of the seven PokeAPI files as checked out (CRLF, Windows; the commit id 7af36d9 is the cross-platform fingerprint), the derived key, and both item sets:

file	sha256 (first 16)
data/raw/pokemon.csv	30f64c395ef64e43
data/raw/pokemon_species.csv	cbf46ea06c6c174a
data/raw/pokemon_types.csv	94a79d654c410263
data/raw/pokemon_types_past.csv	a8f7a684c423d4ff
data/raw/type_efficacy.csv	cc283d6e34641f73
data/raw/type_efficacy_past.csv	e22f07f423f2cb93
data/raw/types.csv	183f4c5a953da62d
data/processed/gen1_key.csv	ccf0cc142b65299f
data/processed/items_s0_n400.csv	ae0c8cd06e0db23e
data/processed/repl_s2_n400.csv	3369976117c667dd

G The ledger

Table 7: **The ledger index.** D: defects in the harness or the plan. O: observations. Full entries, with dates and reproduction commands, in the repository’s FINDINGS.md.

id	subject	finding	status
D-001	known-answer list	six of 26 hand-written expectations were wrong; the derived key was right	fixed
D-002	item sets	PLAN.md named no development set; a 100-item dev set disjoint from the pinned 400 is added for harness work	deviation declared
D-003	prompt and format	PLAN.md did not say whether the model may reason before answering; reasoning is allowed, max_tokens pinned at 1024, parse failures reported as their own metric	decided before any run
D-004	item draw	the pinned set is stratified by answer class, not by attack type: 13 to 45 cells per type, and the four ‘differs’ attack types dominate; the first dev draw had no Bug or Poison attacks at all	pinned set kept, dev redrawn balanced
D-005	scope and budget	PLAN.md’s four conditions, three Anthropic models and \$30 ceiling were written without the multiplication; the budget is \$7 on OpenRouter; one condition (chart + full typing list in context), four models across four labs, a closeness scorer beside exact match; predictions P2, P4, P5, P6 retired and three addendum predictions registered here before any run	deviation declared
D-006	chart format	the markdown table costs 3,032 input tokens per call, not the ~1,600 the addendum assumed, and the dev run shows the table itself is what Haiku misreads; a second chart format (one line per attacking type) is registered as a variant before any pinned run, with prediction A6	deviation declared
D-007	max_tokens	gpt-5-nano hit the 1,024 cap on 30 of 100 dev items with an empty completion: hidden reasoning tokens ate the budget; the 5% rule fired; nano runs at 4,096	fixed, rule applied
D-008	transposition count	the hand count in O-001 (14 of 16) and the code count (10 of 16) use different definitions: any mirrored cell vs the fully mirrored product; both kept, both named	definitions pinned
D-009	upstream routing	OpenRouter split one llama run across two hosts and one of them returned HTTP 500 mid-generation on 31 of 100 items, scored as parse failures; provider pinned, host recorded per run; Qwen was served by ten different hosts in one run	fixed, census added
D-010	model list	the question changed from “four labs” to “how small a model can do this”: a nine-model size ladder replaces the addendum’s list; Haiku and Flash-Lite drop from the pinned run (dev numbers kept); host, quantisation and parameter counts pinned in a registry; A7 registered	deviation declared

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Table 7, continued

id	subject	finding	status
D-011	parse_failures metric	wrong whenever epochs > 1: Inspect’s epoch reduction drops the ‘answer’ field, so the in-log metric reads 0.94 on a run whose true rate is 0.198; analyze, which reads per-epoch samples, is the record	fixed: a ‘parsed’ scorer with a numeric value, negative-tested at epochs=2
D-012	qwen3-32b, table	a thinking model at 1,024 tokens: 14% of table calls truncated before the answer line; rerun at 4,096: 0.828 [0.788, 0.862], 0.08% truncated; the 0.697 floor is superseded	rerun, fixed
D-013	git history	two internal files (the method checklist and a verbatim reviewer text) were purged from history before publication; every commit after the first of them was rewritten, so the commit ids recorded for the addendum, replication and review registrations changed; the blob hashes did not	recorded
D-014	Inspect shuffle	‘multiple_choice(shuffle=True)’ rewrites the logged prompt and completion to look unshuffled and drops the reasoning; only the mapped value survives, so the shown letter cannot be studied from the log; shuffle at dataset load instead	recorded, upstream
D-015	task guard	‘chart=permuted’ was wired through the prompt, the samples and a test, but the task’s argument guard still listed none/gen1/modern; the first REVIEW3 launch failed on every model before any call; guard widened, test added, relaunched	fixed, no cost
D-016	reasoning flag, upstream	REVIEW3.md run 2 asked OpenRouter for ‘reasoning: {enabled: false}’ on qwen3-32b; DeepInfra served reasoning on 1,200 of 1,200 calls (557 output tokens, 12.6% truncated at 1,024); run discarded as registered, X4/X5 unscored; its 0.686 replicates the D-012 floor (0.697)	recorded, \$0.50 spent
D-017	token budget rule	llama-3.2-1b leaves 19.8% of table answers unparsed at 1,024 and was never rerun under the 5% rule; rerun at 4,096: 0.211, 19.3% unparsed, no call at the cap, so the unparsed answers are missing answer lines, not truncation; the registered run stands	resolved
D-018	budget	the 8B’s reasoning traces (1,154 output tokens per call) cost about twice the estimate and the 14B run hit HTTP 402 (credit exhausted) 129 items into its third epoch; reported at two complete epochs plus the partial third, 825 scored calls	recorded
O-001	Haiku 4.5, dev set	84% exact, 0 parse failures; 14 of 16 misses are the chart read transposed (cell (a, b) answered with (b, a)); 0 wrong-Pokemon lookups, 0 multiply errors	observed on dev, not scored against a prediction
O-002	four models, dev set, two chart formats	rows beats table for every model (+0.07 to +0.19 exact); for Haiku and Qwen the transposed misses fall with the format, for Gemini Flash-Lite they do not: its mirror-cell answers come from its prior, not the grid	observed on dev, not scored against a prediction
O-003	two on-device-class models, dev set	llama-3.2-3b at chance (0.18) in both formats; gemma-3-4b below the majority baseline (0.26 table, 0.34 rows) with immunities the worst stratum	observed on dev, not scored against a prediction
O-004	the ladder, pinned set	the knee is between 8B and 12B total params; nothing under 12B beats always-“1”; the MoE sits with its total size; rows beats table for 9 of 10 models, by up to +0.18 after the D-012 rerun (+0.23 before it); predictions scored: P1, P3, A6 (accuracy clause) held; A1, A2 (bucket clause), A4, A5, A6 (transposition clause), A7 (MoE clause) failed	scored
O-005	replication, four knee models, a second 400-item set	order and MoE placement replicate (R2, R3); every model lands inside its pinned interval or within the baseline shift (R4); the “12B beats always-1” half of R1 fails on a set whose majority is 0.478: 12B is where reading starts, not where it beats the dumbest strategy	scored
O-006	rows x3, four knee rungs (REVIEW.md run 1)	the format gain has a band: rows minus table exceeds the sum of both epoch ranges by 2 to 3x for all four (V1 held); every one-epoch rows number within 0.02 of its three-epoch mean (V2 held)	scored

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Table 7, continued

id	subject	finding	status
O-007	qwen3-235b-a22b on a second host (REVIEW.md run 2)	DeepInfra 0.725 [0.679, 0.766] against GMICloud 0.764 [0.720, 0.803]: inside the interval (V3 held, at the edge); the host moves the number by 0.04; on both hosts the 235B MoE is below the 32B dense model (0.828)	scored
O-008	lookup residual and list position (review 2)	below 4B, 15 to 45% of lookup misses state a wrong typing for the defender; from 27B up, 0 to 1%: the small models fail to find the line, the large ones misread the cell; accuracy by list position shows a small primacy effect and no middle dip with no chart and no typing list, recall is 0.14 to 0.28 through 27B and the 30B MoE, about 0.51 at 32B and 235B, 0.665 for the ceiling; the reference lifts 8 of 9 open models beyond the sum of both ranges (W1); below 32B under a third of table hits also come from memory, at 32B and above about two thirds (W2 half held); recall is below the majority baseline through 27B (W3 held)	recorded
O-009	recall pass, all ten models (REVIEW2.md run 1)	under per-item shuffle gemma-3-4b and llama-3.2-3b still answer "2" on 0.63 and 0.48 of items: a value lean, not a letter lean (W4 value clause held; letter clause not measurable, D-014); gemma-3-12b at temperature 0 scores 0.502, range 0.015, inside the default interval (W5 held)	scored
O-010	shuffled options and temperature 0 (REVIEW2.md runs 2, 3)	the two Qwen MoEs are the -2507 instruct variants: zero reasoning blocks, zero truncation, ~167 output tokens per call; qwen3-32b reasoned on every call (~618 tokens); the D-012 artefact does not touch the MoEs, but the 30B-A3B vs 32B gap is sparse-vs-dense and thinking-vs-not together; REVIEW3.md run 2 separates them	scored
O-011	the Qwen rungs, thinking mode (review 3)	with the 15 type names permuted (memory of the real chart useless), every model lands within 0.06 of its real-chart score (X1 5 of 7; X2 failed: the 235B and the ceiling moved under 0.02, the 32B's drop is its 1,024-token budget); the contradicted cells rise from 0.43-0.60 to 0.52-0.82 from 12B up: the prior is a tax on the cells it contradicts, not a subsidy on the rest	scored
O-012	relabelled chart, all ten models (REVIEW3.md run 1)	paired over items: Gemma 4B->12B +0.233 [+0.185, +0.279]; Llama 1B->8B +0.057 [+0.019, +0.092] (p=0.007, so not inside the noise as the draft said); 27B vs 30B-A3B +0.026 [-0.014, +0.068] (level); rows gain excludes zero for 9 of 10; within answer class the contradicted-cell gap is clear from 27B up and inconsistent below; with immunity filed before mirrored the immunity share is flat 0.13 to 0.24 at every rung but 235B	scored
O-013	paired tests, within-class differs gap, taxonomy ordering (review 4)	on the 329 cells where the charts agree, relabelling costs the 235B 0.040 and the ceiling 0.029 (27B 0.000, 30B-A3B +0.024); the prior is a tax of 0.07 to 0.22 on contradicted cells and a subsidy of at most 0.04 elsewhere, X2's direction under its 0.05 threshold; the rows prompt is 6% longer than the table prompt (2,891 vs 2,728 tokens on Gemma)	scored
O-014	relabelled chart, the agreeing cells (review 6)	qwen3-8b 0.850 [0.81, 0.88] and qwen3-14b 0.896 [0.85, 0.93] under table with reasoning on, against 0.828 for the 32B, 0.642 for the 30B-A3B and 0.279 for llama-8b: a thinking 8B beats every non-thinking model on the ladder; the knee is a one-pass-answer property; Y1 and Y2 and Y4 failed, Y3 held	scored
O-015	Qwen3 dense 8B and 14B, thinking on (REVIEW4.md)		